Ex:-7:- On which dates of April, 2021 Sundays falls?

Ans:- To find out Sundays (or) any day of week on the given month of given particular year, first we need to know 1st week day of that month.

01-04-2021 → day

$2000 + [20 + 5] + \underbrace{3+0+3+1}_{0 \text{ odd days}} = 4$ odd days $\Rightarrow \therefore$ Thursday.

[0] ; 4 odd

First Sunday is on 1st-04-2021. Thursday + 3 days.

$= 04^{th}$-04-2021.

Next sundays with a gap of 7 days are 11, 18, 25

So, Sundays on $\boxed{4^{th}, 11^{th}, 18^{th} \text{ & } 25^{th}}$ of April, 2021.

Ex:-8:- The next leap year to 2896th year?

Ans:- generally 2896+4 = 2900 is leap but being a century not divisible with 400 it's not a leap year.

So, 2900 + 4 years = 2904 is next leap year to 2896.

Finding same calendar year:-

Note:-1:- If the given year is Leap year then same calendar year after 28 years.

Note 2:- If the given year is like L.Y+1 (or) L.Y-3 then same calendar year after 6 years.

Note 3:- If the given year is like L.Y+2 (or) L.Y-2 then same calendar year after 11 years.

Note 4:- If the given year is like L.Y+3 (or) L.Y-1 then same calendar year after 11 years.

* Note 5:- If the given year is like [Non-leap century year - 28] years ~~then same~~ which is leap year then same calendar year is not after 28 years. but after 40 years same calendar year will come.

Ex:9:- What will be the same calendar year of 2021?

M-1:-

1	1	1	2	1	1	1	1	1.
2021	22	23	24	25	26	27	28	29 . . .

always after 7 multiple number of days same day is repeating.

∴ from 2021 → 1 odd day to 2026 → 1 odd day the odd days count is '7' then next year 2027 starts with same day. It ~~year~~ is same calendar year. if given year is N.L.Y/L.Y then it's also N.L.Y/L.Y. if not go through the same counting of odd days till to get same calendar year. So 2027 is same year.

M-2:- 01-01-2021 → 5 odd days ⇒ Friday.

01-01-2022 → Saturday (+1) → odd day in 2022

01-01-2023 → Sunday (+1)

01-01-2024 → Monday (+1)

01-01-2025 → +2 → ~~Wednesday~~ Tuesday

01-01-2026 → ~~Thursday~~ Thursday (+1)

01-01-2027 → Friday So, 2027

Rt-3:- By Note point given year is like L.Y+1 / L.Y-3

∴ same year after 6 years.

∴ 2021+6 = 2027 is the answer.

Ex:-10:- Which year will have the same calendar year of 2172 ?

Ans:- 2172 looks like leap year then generally same calendar year after 28 years. Then it is

2172+28 = 2200 [being a century year to become leap year it must divided by 400 but 2200 is not divisible by 400. So, it is non leap year].

In this exception. given year looks like [non-leap century year-28] then same calendar year after 40 years to the given year.

Answer is 2172+40 = 2212